

I: Alright so I think now it's recording. I want to start by saying that teaching and learning have been affected in the last few months because of the pandemic so in any of the questions I am going to ask you, please feel free to reference your normal teaching practice or even considering how you adapted your practice for the situation, or both. So, if you want to give examples and explain how you did things prior to Covid 19 that's fine or if you want to reference your practices before March, that's fine as well. Okay.

I: So, the first question is could you briefly describe your experience teaching undergrads like how does your teaching relate to your past research or current research, could you describe the courses where you teach students to work with data.

UCSB08: Sure, so I teach a statistics course for the [department redacted] in the [level redacted]. So, most of the students are probably in their sophomore year, transfer students probably in their junior year. And it's actually the last class that students have to take before they're allowed into the major. So, I call it a high stakes class so they have to make a certain grade in the class, usually in order to enter this like the [major redacted]. So, it is a statistics class. So, in that sense it's grounded in a lot of data. And we cover statistical concepts both conceptually as well as we do hand calculations of statistical tests, as well as the course has a lab section, and in the lab section the students are shown how to conduct statistical analyses in R statistical software. So, it varies from quarter to quarter. I have anywhere from 125 to 250 students in the class...so, this year is a little bit smaller. And I usually have 6 to 10 TAs helping with the lab sections. And the lab sections are set up for students. We give them a data set and they work their way through statistical analyses and then they report those analyses in APA style. I wouldn't say that I linked the class directly to my own research per se, so I've set it up a couple of different ways. So previously I had given them fictitious databases to work with. So just kind of data that I made up and simulated to work with but this quarter, what I actually did is I sourced data from faculty and [department redacted]. So, they actually gave me data from published papers that they had. And the students are running statistical analyses with their data, with faculty members data essentially. So, it's nice to ground the data analyses back into some real world, [discipline redacted] data collections that have happened within the department.

I: Great. So that kind of ties to my second question which relates to how students usually get data. So, you mentioned that you usually provide them these data sets, right, based on publication.

UCSB08: For this class, yes.

I: Okay, are all their sources of data sets, like did you have any assignments for instance that the students were encouraged to find a given data set, find like primary resources to conduct some of the assignments they have or will just say, this is usually your primary method

UCSB08: Previously I've only given them data sets. The reason being is because they have a weekly assignment that's do every week and it's really intensive grading for the teaching assistants and so it's much, I think, easier for the teaching assistants, as well as probably fair for

the students for all students to be conducting the same analyses with the same data so that the TAs know what to look for when they're grading those assignments and it kind of puts everybody on an equal playing field with regards to the process of analyses. I can certainly see a situation where it would be useful to have students go out and source their own data. But for this class because it's a pre [major redacted] for example we don't have them collecting any of their own data, they do that in upper division classes after they're in the [major redacted]. But what we're trying to do in this class is really just to get them comfortable with statistics and, you know, using R for the first time, those sorts of things. So, it's more of a, I mean they have a stats class before they take my class. I believe that class I think it's [course code redacted] or something like that focuses a lot on hand calculations and probability. And so, what we do in this class is try, what I do in this class is try to make it a bit more conceptual and grounded in [field redacted] data, and the ways that we use statistics in our fields so I really try to expose them to the specific analyses that we tend to use a lot. And the way that we would write up and interpret those results in APA style.

I: And in terms of, like, providing access to these data sets. What are the most common approaches you follow like to provide your students with the data sets? Do you share through Gauchospace or email? How do you usually make these data sets available for your students, and the TAs that are helping you.

UCSB08: So, usually, I post the data sets up on Gauchospace. So, for example, I posted one up on the first week of this quarter and they've had a reading associated with it so the students read the empirical paper, I posted the data from the faculty member within the [major redacted], and the students have worked with that data, the same data set for a couple of weeks so running some descriptive statistics and T tests some one way ANOVAs. And now we'll post up a second data set that the students will use for a few more lectures with an associated paper with it to run some correlations and regressions. So, usually, I just give it to them through Gauchospace.

I: And do face any challenges in finding or obtaining data sets for teaching, or would you say this is relatively easy in your field?

UCSB08: I think it's pretty easy. I mean like I said, in the past I've made up some data and sometimes I've used some of my own data prior to teaching here at UCSB. But this quarter I approached faculty members and said, this is the idea that I have, I'd like my students to work with real [major redacted] data that faculty in our department have actually used and published with, can I have access to your data? And everybody was more than willing to share it, which was nice. I thought about using like national data sets that are available, but I think the number of variables and the number of cases would be really overwhelming for first time R users to open up a data set and see you know 1000 variables and 10,000 participants would be a little bit much I think.

I: Okay. So we talked a little bit about getting access to data or providing access to data sets. Now I would like to shift my questions a little bit more about the working component so working

with data. So how would you say your students manipulate, analyze, and interpret data in your courses and classes? You have provided descriptions that they are using R mostly right for this class to run some statistical tests. Could you describe a little bit more about this process, how they work, or give an example based on one of the assignments you have for the classes you teach.

UCSB08: Sure. So, we actually use data in two ways. So, in the lectures I actually walked through how to hand calculate statistics and more of the conceptual background for a particular statistic like a T test for example. And I also, when was that, not this past summer but the summer before, created a series of learning glass videos. I'm not sure if you've seen learning glass before. It's like a one way or two way. Right so basically I created a series of, I call them stats class videos to show students how to hand calculate statistics. So, for example, in one week we might work with some conceptual ideas of what's a T test, and then I'll show them how to hand calculate a T test and these stats class videos and then then the lab section that week, they would be given a data set like we just talked about, and they would learn how to use R. The TAs would walk them through the code that's used in R in order to run a T test with the dataset that they have. And in both cases we focus a lot on interpreting the outputs of R or the outputs of our hand calculations to determine whether or not results are statistically significant using null hypothesis statistical testing so we focus a lot on that in the class. And then there's a big emphasis like I said on just writing it up correctly, so I tend to err on the side of I really want students to be able to not just calculate statistics and values but to understand what those values mean on the other side, and how to report those values to someone else because I think that helps them with actually reading empirical papers. So, when they practice writing a results section I think they are better equipped to actually read the results section of the paper.

I: Right. And so you mentioned R, are there other tools they use during your classes or would you say it's mostly R?

UCSB08: We use Excel sometimes to help with graphs and developing charts and graphs and line graphs and so forth, but mostly we use R. Previous to coming to UCSB, at my previous university I taught SPSS to students but since being here, the focus has been on using R.

I: Okay. Yeah. And so you mentioned they have a previous statistics class, so that they learn how to hand calculate and also go through some of the tests like T tests and ANOVA in R. For this case, would you say they have any prior knowledge. Is there any prior knowledge you'd say, of tools or software do you expect our students to have before taking your courses or. You mentioned TAs are teaching and walking them through R, but then or, like, do they learn all the skills they need for this course throughout the course?

UCSB08: I think when it comes to R they learn everything in the lab section. So, I think they learn everything in this course, my understanding is that they haven't had that exposure in a previous stats class. In fact, I've been quite surprised, for example even working with Excel some students have a difficult time and don't seem to have much experience in working with Excel or uploading something to R for example. I mean it requires a little bit of coding

knowledge but the process is kind of the similar, you know, similar to uploading a photo to to an email, but I do find that when it comes to statistics for whatever reason students don't think of the process as the same sometimes and so yeah I do feel like everything we teach them is pretty much brand new when it comes to actually using R to analyze their data, I mean the hand calculations and the conceptual ideas, some of that they've had before. I've never seen the like syllabus from the previous stats class that the students have to take before they get into my class. I've tried to access it a couple times. I think I saw an abbreviated version of one once. It's my understanding in the first class that they focus on some statistics in depth that we don't use as often in [major redacted] so they might focus a lot on Chi squares and we don't use that a ton and so I try to focus on the analyses that we use a lot. So, T tests, ANOVAs and regression and correlation for example, we might dabble in chi square at the end of the quarter if we have time but we don't spend a lot of time on it because in [field redacted], we use a lot of the other analyses. But I do actually feel like even with the conceptual and hand calculations they're learning or relearning a lot of it in my class.

I: Okay. And do you face any challenges relating to students' abilities to work with data in your classes because you mentioned that you want them to go beyond just calculating but like be more critical about statistics. What would you say could be challenges for your students to work with data.

UCSB08: I think, well one of them I think is for students to move beyond thinking that data and statistics is just math. So, I think that's a hurdle that I try to overcome quite often to get a sense of it's more than math and that's out there in everything we see. So, there is a statistic on everything, perhaps, especially right now. So, there's lots of statistics out there and then it's up to us to understand and interpret them correctly, not necessarily up to the person presenting them sometimes. So, I do think that that's a big struggle is to move beyond statistics equals math to statistics is more than that. That data is more than that, that data can help us better understand things around us in terms of the things that we see in our everyday life. So, you know if they know how to read a table better from taking the stats class then we'll be able to use that in all sorts of situations. So, I think that's definitely one challenge. I do think that there's a general anxiety associated with data among students. So, like I mentioned before, I feel like I was like, upload a, you know, a photo onto an email, they know what I mean when I say upload. But when I say it in the context of "upload your data set into R" it feels very novel and foreign and there's a sense of anxiety associated with it sometimes. So, yeah, I think that's part of it as well as overcoming that kind of anxiety that students might have about interacting with data, and that when they push that little Run button, the whole thing's not going to melt in front of them.

I: Right. So, in terms of your field, how do the ways in which you teach with data relate to the learning goals in your field? What will be your opinion on how it connects to the approaches that are taken in [field redacted]?

UCSB08: Can you repeat the question?

I: Yeah, so how do the ways in which you teach with data relate to the learning goals in your

field?

UCSB08: Well, I think, I mean thinking critically as a big one in [major redacted] and I think [major redacted] is, I mean it's science and so I think any science field has data associated with it. That might be a hurdle that we have to overcome in [major redacted] where it's thought, [department redacted] particularly, that it's thought that perhaps [major redacted] is just interacting with patients. You know, so a lot of students think it's about being [profession redacted], and overcoming that idea and helping students recognize [major redacted] as a science, dealing with data, multiple types of data different types of data from humans and animal subjects and so forth, is important, and aligned with the goals of [department redacted].

I: Right. So you mentioned about teaching students to think critically about the sources and the uses of data they may encounter like in papers. Could you describe, like, a situation or if you feel you have an anecdote or like a situation, actually, you had an assignment where students had to go and think critically about the data source they were using. Could you talk a little bit more about this. Like how do your students think critically about the data source they're using?

UCSB08: I think the critical thinking part in my class comes with the interpretation so everybody, if everybody's running the same analysis, for example, everybody's running a T test for the same data set everybody gets the same output. But that doesn't necessarily mean that everybody's going to write it up in the same way. So, I think in this course in particular critical thinking comes with considering what the values that they get on their output mean in the context of the example study that they have. And everybody writes differently and I think that's the nice thing about having this writing component in the stats class is that you get more than just calculations and values you have students interpreting the data. Yeah, I think that comes with thinking critically. I think just in general, in order to conceptually understand statistics there has to be a component of critical thinking and I try to walk students through that process in my lectures in particular, so the lab sections focus a lot on R, and in my lectures I focus a lot on the conceptual understanding of concepts. And so sometimes I'll have, I don't know, three or four lectures where we don't even calculate anything. We just talked about some of the concepts of statistics.

I: Right. So, if you could list out a few specific data skills that you believe you prepare your students, in terms of future careers. What would be those?

UCSB08: Like specifically for future careers?

I: Yeah.

UCSB08: So yeah, so, there's a couple skills, one is being able to see data represented visually so graphs, tables, charts, and being able to interpret that data is important. I also try to teach students that the onus is on them if they're the researcher to present the data accurately. So we talk a lot about that in the quarter how you can, you know, change the y axis on a graph and make an effect look bigger than it actually is for example and that the onus is on the researcher,

the data collector to present that accurately. And in doing so I hope that means that they question the data that they see out there in the real world. Yeah. So, I think those are some skills that I think would go forward. I mean, coding is a big thing now for I think a lot of professions. But I'm not trying to get my students to leave my class experts in R, they get it for nine weeks, and, you know, they'll probably look up whatever code they need to use again in the future. I don't make them memorize the coding for the R for example in this class. But I do think at least they get some exposure to coding which is important for their future.

I: Alright, the next questions about your perception in terms of cultural changes at UCSB or even outside UCSB in terms of the ways in which instructors that are teaching with data. So, have you observed any policies or cultural changes at UCSB or outside that are influencing the way in which you teach with data?

UCSB08: I don't know if they're changing so much the way, I mean previously like I mentioned I was using made up data sets about kind of silly [field redacted] related examples, perhaps. But this quarter moving to using real [field redacted] data sets, I did intentionally for example choose a data set that looked at cultural differences in the concept that will start working with the week after next in the class so we will have a data set that specifically focuses on cultural differences and to explore that in a data analysis. But, yeah, so I think that's part of it.

I: Yeah. Okay. Thank you. The next question is about your perception in terms of ethical challenges and teaching with data. So, do you see that instructors in your field face any ethical challenges in teaching with data? And, if so, to what extent are these challenges important to your teaching?

UCSB08: I mean, speaking for myself, because I don't have students go out and collect data, it's very different so they're not interacting with participants, they're not asking each other questions. They're working with previously collected data that has all been de identified where there's no identifying information about who the participants were. Everything was confidential. So, in the statistics class that I teach we don't have to deal with that too much. However, sometimes I teach [course number], once a year usually, which is a research methods class. And in that class we talk about ethical principles in [major redacted] and how to follow those principles when conducting research and collecting data, but not necessarily in the stats class that I'm teaching this quarter.

I: OK, so now the questions are going to move to training and support. So, you mentioned that you have help from some TAs, correct, in this class, that are particularly involved in teaching R coding to your students. But in your course, would you say that you have any other type of support, in terms of helping to provide instruction and support for your students to, you know, obtaining or even working with data, rather than the students you already mentioned. So, for instance if you have help from librarians or you have drop in sessions, like from different divisions or like different departments, coming in and kind of doing demos or getting involved in your instruction.

UCSB08: Not so much in this class because it really is an introduction to statistics. So, it is kind of a baseline level class and because I have so many TAs. There's quite a bit of knowledge among the graduate students from [department redacted] about using R. I mean I have relied on tech support every once in a while, particularly you know during the pandemic where not all students are able to download R onto their computers or I think it doesn't work on, doesn't work on a particular type of iPad, I can't remember what, but we have access to R cloud. So, the students have been able to use R cloud in the class, I think there's about five students out of 125 this quarter that are using it. But yeah, I don't have any drop in lectures. And that's probably it. They use Gauchospace a lot to communicate with students in terms of providing the data, providing lab slides for writing lecture slides, send you rely on that quite a bit, providing the, I mean the stats class videos I have. I recorded those over so I had a lot of support with getting those done.

I: Great. So, to your knowledge, are there any ways in which your students are learning to work with data outside their formal coursework. So, for example online tutorials, internships, peers. Are you knowledgeable about these alternative ways they're learning how to work with data?

UCSB08: I think there's well in the [department redacted] students can be research assistants, undergrad students can be research assistants and a faculty members lab. And that seems to me to be the next place where they get some additional experience outside of a formal class setting, I mean they are registered for it through a credit, but not in a formal way where they're receiving lectures and so forth. But yeah, I think most students probably have some experience and exposure, who are research assistants to data in that way, whether it's through data collection or through analyzing data and working with coding data.

I: And for your classes, do you have any type of, or do you expect or even encourage this kind of extra-curricular learning. So, do you have any, like, additional resources that encourage your students to go forward and like to learn more about the, not only the statistical approaches or analysis but also like the tools that are available.

UCSB08: No, haven't used that much in my class. I mean, I feel like the classes so packed with work for the students that I'm not sure even if I recommended something like that, particularly now I think the students are totally overwhelmed with taking online classes so I mean I feel like even if I did recommend it, I'm not sure the extent to which a lot of the students would use it. There'd probably be a handful that might, but I wouldn't want to put it out there and make students feel required to use it at this point so, anyway. I try to make things a little easier, not easier for them but less, less intense for them at the moment.

I: Yeah. Makes sense. Yeah. So, I'd like to hear a little bit about your own training. So have you received any training teaching with data other than your graduate degree or you mentioned you had a past job in another institution. But let's say like, workshops, technical support or help from peers, how did you get your experience in teaching with data, how did you learn about like what would be good approaches to teach with data?

UCSB08: I think it probably started in graduate school at some really excellent statistics teachers in graduate school and when I finished my PhD. I can't remember what it was, I had taken so many stats classes that I had some sort of certificate in social science methodology and statistics. And then, I usually do when I attend conferences, if there's some novel statistical concept that I'm interested in but not familiar with, I often attend a workshop, I think, pretty much every conference I go to I try to do that, to attend a workshop at a conference about some novel statistics. I've attended some talks at the Q, the quantitative QMMS series. So, I've given a talk there and then I've also attended a few talks so on didactic analyses which I wasn't so familiar with. And then prior to coming at UCSB I was teaching statistics and methods at undergrad and graduate level for 10 years in the [region redacted], so my previous university, [institution name redacted], and [country redacted], and there that's all I taught for 10 years was statistics and methods at both undergrad and graduate level.

I: So you mentioned that you upload these data sets, mostly to Gauchospace, that's like your environment actually students get access to these data sets. Do you use any data sets, some data sets you mentioned already, but like assignment plans, syllabi, or other instructional resources that you have received from others or from your peers? And how do you make available, or do you make them available, if so, how do you make available your own resources to others. And when I mention others it's like outside your classroom so not only your students, but also your peers.

UCSB08: Yeah, so maybe, maybe not so much, I'm trying to think. So, when I first got to UCSB, when I first came to UCSB I did borrow resources from faculty in my department who had previously taught the course just to get a sense of the structure of the course kind of what students might be expecting what the department was sort of expecting with how the course was formatted and run. So, I did work with those previous lectures and assignment ideas and tests and so forth as a starting block. And then I've also shared my resources with graduate students who have taught the class in our departments, they usually teach in the summer. And then, outside of the department I'm trying to think if I've shared materials from this class. I did, I can't remember if it was this class or different class but I think it was this class, yeah it was this class. So, the APA American Psychological Association has a syllabus resource center for faculty teaching methods and statistics and so I did share my syllabus there as part of the, I guess syllabus bank that they're developing. Yeah, that's kind of the extent of it yeah.

I: Yeah, great! APA.

UCSB08: Oh, and my stats class videos I offered those all to my colleagues to use.

I: Right, so there's like a lot of sharing in your community.

UCSB08: Yeah, in our department there seems to be.

I: In terms of, like, considering the evolving trends in your field, what types of training or assistance will be most beneficial to instructors in teaching with data? So, in your opinion what would be like beneficial for instructors to learn, to get prepared in teaching data.

UCSB08: Um, well I think, I don't know how other fields are but in [field redacted], I know that R is the fastest growing software currently, and I wasn't trained in R, so I was trained a long time ago and I was trained in SAS and then I was trained in SPSS. And so actually I had never used R until I came to UCSB to teach it, which was pretty interesting. So, I was kind of learning R just before I was teaching it to students and setting up lab sections to teach it. So, I think that would have been a bit useful for me looking back to have a, you know, as a new faculty member to not feel kind of self-conscious about not knowing the newest statistical software out there and have some resources to learn that would have been helpful. I mean, I feel like I have the hang of it now, but it would have been nice early on to get that sort of resource. I think that's the only thing.

I: Yeah. Do you see any other skills that will be important for instructors to learn or focus on before teaching with data.

UCSB08: I mean, I think maybe some, I mean I've been teaching it so long that I feel like it comes kind of second nature. But I remember initially when I first started teaching statistics I didn't realize how tentative students were about working with data and learning statistical concepts, that kind of, I think that sort of anxiety that students had and the fear that they had of learning a new statistical software program and learning about data and what data can tell you, I was kind of shocked by that sort of anxiety that they had. And so, it might have been nice to kind of talk about that with colleagues ahead of time and help manage that with students.

I: Because it feels like it's all at once, right? They're new

UCSB08: And they're all so anxious. Yeah.

I: Alright, so my last question to wrap up, I would like to ask if there's anything else from your experiences or perspectives as an instructor or on the topic of teaching with data more broadly that he would like to share.

UCSB08: I don't think so. I mean, I think, in our field, [department redacted], I think data is used in almost every class that's taught. So, I do think that our students get a lot of exposure to it, which I think is great. And like I say, for me, not all [major redacted] majors are going to go on to do something in [major redacted] so it's more important for me, that they can appreciate other people's presentations of data and kind of weed out when it seems legitimate versus maybe when it doesn't. So, for me, that seems like a really important skill for students to leave with having that sense of feeling comfortable with seeing data, seeing graphs, hearing somebody say a percentage of people did x y and z and knowing what that means. It feels important.

I: Great, so I'm going to stop recording.